

Education

Yale University

Candidate for BS in Mechanical Engineering (ABET), Varsity Athlete (Crew)
Noble and Greenough School
Highest Distinction

2024 - Present

GPA: 4.0

2018 - 2024

GPA: 11.0/11.0

Skills

Design & Analysis: NX, Teamcenter, SolidWorks, 3DX, OnShape, GD&T, ANSYS Workbench, Mechanical, Fluent, CFX

Programing: Python, Java, VBA, Excel API, SolidWorks API, MATLAB, Arduino, Ardupilot, Px4

Manufacturing: Lathe, Mill, Shop machines, Power tools, Soldering, FDM, SLA, SLM, MJF printing

Work Experience

SPACEX | Starbase, TX

June 2026 - September 2026

Booster Vehicle Design Intern

- Upgraded booster capability and reliability while prioritizing production at rate, releasing 40+ ECNs.
- Designed critical high-performance sheet metal joiners and stringers for B23+ tank gross acreage, fuel tank door, and grid fins, working with analysis teams to reduce shear, buckling, and thermal failure modes.
- Coordinated with propulsion and structures teams to integrate raceway DFM and capability upgrades across all tank stations, including ownership of high level assembly releases.

DRAPER | Boston, MA

June 2025 - August 2025

Mechanism and Electromechanical Design Intern

- Supported electronics development and testing for defense applications.
- Automated creation of precision cut and form tooling for quick-turn IC development using parametric modeling and Solidworks API, resulting in auto-generated models and GD&T drawings.
- Developed cooling system data infrastructure and control software for Celestial Navigation System (CNS) using TCP and Modbus. Used on next generation aircraft.
- Designed and analyzed mass model of CNS system to meet prime contractor requirements, performed fatigue, shock response spectrum, modal, and functional shock analysis using ANSYS and SolidWorks.

UNITZERO | New Haven, CT

December 2025 - June 2026

Robotics Consultant

- Designed Universal Manipulator Interface (UMI) with Camera, VR tracker, and Jetson Nano for Vision-Language-Action (VLA) model training.
- Designed 3D-printed 6-DOF hall-effect based leader arm for robot arm control and data collection.

Project Experience

YALE UNDERGRADUATE AEROSPACE ASSOCIATION | New Haven, CT

Project Lead, Yale Jet Aeronautics

January 2025 - Present

- Attempting to break the sound barrier with an air-breathing vehicle.
- Leading the design, manufacturing, and testing of 350N afterburning jet engine and airframe.
- Performed Turbomachinery design and optimization using CFX.

Aerostructures Lead, Project Liquid

September 2024 - June 2026

- Responsible for airframe design, packaging, and regeneratively cooled liquid rocket, subteam leadership.

FIRST ROBOTICS COMPETITION TEAM 125 - NUTRONS | Revere, MA

December 2017 - June 2024

Team Captain, Team Leader of Design and Systems Integration, Subsystem Design Lead

- Led team to finals of world championship, 12 event wins. Designed launchers, feeders, telescoping arms.
- Owned overall robot integration, coordination between hardware, software, and manufacturing subteams.

PERSONAL PROJECTS

Hybrid VTOL

August 2023 - December 2024

- Designed and built patent-pending wingeron/tricopter hybrid capable of plane-like forward flight, efficient hover, and decoupled pitch & translation. Successfully evaluated forward and tricopter flight modes.
- Wrote custom flightcontrol software for Teensy microcontroller.

High-Efficiency FPV Racing Wing

October 2023 - January 2024

- Designed, optimized, and built FPV racing wing capable of 102mph flight on only 3s LiPo powertrain.
- Used Xflr5 for airfoil, sweep, and wash optimization, longitudinal stability analysis.

Extracurricular Activities and Awards

- National Merit Scholarship Finalist 2024
- First Robotics Einstein World Championship Finalists 2023
- Team Captain, Noble and Greenough Boys Varsity Crew 2024